

Stroke (Brain Attack) — Comprehensive Clinical Guide

1. Definition and Epidemiology

Stroke is defined as a sudden neurological deficit attributable to focal brain, spinal cord, or retinal ischemia or haemorrhage, with imaging or pathological evidence. A Transient Ischemic Attack (TIA) has the same mechanism but symptoms resolve within 24 hours (usually within 1 hour) without infarction on diffusion MRI.

Stroke is the second leading cause of death globally and the leading cause of adult disability. In India, the incidence of stroke is 105-152/100,000/year, with approximately 1.8 million new strokes occurring annually. India is witnessing increasing stroke burden in younger adults due to hypertension, diabetes, and atrial fibrillation.

Time is Brain: every minute a large vessel occlusion stroke goes untreated, approximately 1.9 million neurons, 14 billion synapses, and 12 km of axons are destroyed. Rapid recognition and treatment is the single most important determinant of outcome.

2. Classification and Mechanisms

Ischemic stroke (87% of all strokes): caused by focal brain ischemia due to vascular occlusion. TOAST Classification: Large artery atherothrombosis (carotid/vertebrobasilar), Cardioembolic (atrial fibrillation, valvular disease, intracardiac thrombus), Small vessel lacunar (lipohyalinosis of perforating arteries in hypertension), Other determined cause, Undetermined (cryptogenic).

Haemorrhagic stroke (13%): Intracerebral haemorrhage (ICH, 10%) — spontaneous bleeding into brain parenchyma, most commonly from rupture of small penetrating arteries damaged by chronic hypertension. Subarachnoid haemorrhage (SAH, 3%) — bleeding into subarachnoid space, usually from ruptured intracranial aneurysm or AVM.

Large vessel occlusion (LVO): occlusion of proximal cerebral arteries (ICA, M1/M2 MCA, basilar artery). Responsible for 20-25% of ischemic strokes but ~50% of stroke disability. Clinically characterized by devastating deficits and amenable to mechanical thrombectomy.

3. Stroke Recognition — FAST and Beyond

FAST: Face drooping (unilateral facial droop, asymmetric smile), Arm weakness (one arm drifts down when both raised), Speech disturbance (slurred, garbled, or unable to speak/understand), Time to call 112/911.

BE-FAST (more sensitive for posterior circulation stroke): Balance loss (sudden), Eyes (sudden vision change in one or both), Face, Arms, Speech, Time.

Other stroke symptoms: sudden severe headache 'worst headache of life' (SAH — thunderclap headache), sudden loss of vision in one eye (amaurosis fugax — retinal artery occlusion), sudden vertigo with ataxia (posterior fossa ischemia), confusion, memory loss.

Stroke mimics: hypoglycemia (most important — always check blood sugar), Todd's palsy (post-seizure weakness), complex migraine, hypertensive encephalopathy, brain tumour, functional neurological disorder, Wernicke's encephalopathy.

4. Pre-Hospital and Emergency Management

CALL 112/911 immediately. Do NOT drive patient to hospital. Ambulance is critical for pre-notification and direct access to stroke team.

In-ambulance assessment: FAST, blood glucose, time of symptom onset, airway-breathing-circulation, IV access, ECG. Pre-notify stroke centre: 'STEMI for the brain' — activate Stroke Code.

CT brain (non-contrast) within 25 minutes of arrival: rules out haemorrhage (contraindication to thrombolysis), detects early ischemic changes (ASPECTS score). CT angiography: detects LVO for thrombectomy consideration. CT perfusion: identifies salvageable penumbra.

IV thrombolysis with alteplase (tPA): 0.9 mg/kg IV (max 90 mg, 10% as bolus, rest over 60 min) within 4.5 hours of symptom onset for eligible ischemic stroke patients. Not for haemorrhagic stroke, recent surgery, anticoagulation, severe hypertension (>185/110 — must be controlled first), or recent stroke.

5. Mechanical Thrombectomy

Mechanical thrombectomy (MT) is the most significant advancement in stroke treatment of the 21st century. Stent-retriever and aspiration catheters physically remove clots from proximal cerebral vessels under fluoroscopic guidance.

Indication: LVO ischemic stroke, within 6-24 hours of last known well, with salvageable tissue on imaging (DAWN, DEFUSE-3 criteria). mTICI 2b-3 reperfusion achieved in >80% of cases at experienced centres.

Outcomes: MT increases the proportion achieving functional independence (mRS 0-2) by ~25% vs medical treatment alone. A number needed to treat (NNT) of ~2.6 makes MT one of the most effective interventions in medicine.

In India, MT capability is concentrated in major urban centres (AIIMS, CMC Vellore, Apollo, Fortis, PGI Chandigarh). Expanding to tier-2 cities is a priority. Telestroke programmes can extend expert evaluation to district hospitals.

6. Secondary Prevention of Stroke

Antiplatelets: aspirin 75-150 mg daily + clopidogrel 75 mg for 21 days (high-risk TIA and minor stroke — POINT/CHANCE trial), then aspirin alone. Ticagrelor + aspirin being evaluated. For cardioembolic stroke — anticoagulation, not antiplatelet.

Atrial fibrillation and stroke: AF is the most important modifiable risk factor for cardioembolic stroke. CHA₂DS₂-VASc score ≥ 2 in men, ≥ 3 in women → oral anticoagulation (NOAC preferred over warfarin: apixaban, rivaroxaban, dabigatran — lower bleeding risk).

Blood pressure management: target <130/80 mmHg. Start antihypertensive treatment 24-48 hours after stroke in stable patients. ACE inhibitor + diuretic (perindopril + indapamide, PROGRESS trial) reduces recurrence by 43%.

Statin therapy: high-intensity statin (atorvastatin 40-80 mg) regardless of baseline cholesterol for all ischemic stroke/TIA patients. Reduces stroke recurrence and coronary events.

Carotid artery stenosis: symptomatic carotid stenosis >50% → carotid endarterectomy (CEA) within 2 weeks (best within 48 hours for high-risk TIA). CAS (carotid artery stenting) for high surgical risk patients.

7. Haemorrhagic Stroke Management

Intracerebral Haemorrhage (ICH): acute BP lowering to SBP <140 mmHg is recommended (INTERACT2, ATACH-2 partial). Reverse anticoagulation: vitamin K + FFP/PCC for warfarin; idarucizumab for dabigatran; andexanet alfa for Xa inhibitors.

Surgical indications for ICH: cerebellar haematoma >3 cm or causing brainstem compression (urgent surgery); lobar ICH with deterioration. Basal ganglia and thalamic haematomas — no surgical benefit in most cases. Minimally invasive surgery (MIS) for ICH under trial.

Subarachnoid Haemorrhage (SAH): immediate CT (100% sensitive in 12 hours of ictus), then lumbar puncture if CT negative but thunderclap headache strong clinically. Secure aneurysm: coiling (endovascular) or clipping (surgical) urgently. IV nimodipine 60 mg every 4 hours for 21 days to prevent vasospasm.

8. Stroke Rehabilitation

Rehabilitation begins within 24-48 hours of stroke stabilization. Early mobilization reduces pneumonia, DVT, pressure sores, and depression. Stroke unit care (vs general ward) reduces death and dependency by ~25%.

Multidisciplinary stroke team: physiotherapy (motor recovery, gait training, balance), occupational therapy (ADL retraining, hand function, adaptive equipment), speech-language therapy (dysphasia, aphasia, dysphagia), neuropsychology (cognitive rehabilitation, depression, anxiety).

Principles of neuroplasticity-based rehabilitation: task-specific repetitive practice drives cortical reorganization. Constraint-induced movement therapy (CIMT), robot-assisted therapy, virtual reality, non-invasive brain stimulation (rTMS, tDCS) are emerging modalities.

Aphasia after stroke: 30% of stroke patients. Language rehabilitation (speech therapy) should be intensive and prolonged. Communication aids, family education, aphasia groups, and community reintegration support.

9. TIA and Stroke Prevention

TIA is a medical emergency. 10-15% risk of early stroke after TIA — ABCD2 score (Age, BP, Clinical features, Duration, Diabetes) helps risk-stratify. High-risk TIA (ABCD2 ≥ 4 or symptomatic carotid stenosis or AF) → immediate hospitalization, urgent evaluation, and

treatment within 24 hours.

Brain and vascular imaging: DWI-MRI is more sensitive than CT for acute ischemia. MRA or CTA to assess intracranial vessels. Carotid ultrasound and Doppler for carotid stenosis.

Cardiac investigations: ECG (AF), 24-hour Holter, echocardiogram (thrombus, valvular disease, patent foramen ovale).

Long-term risk factor control: The single most preventable risk factor for stroke in India is uncontrolled hypertension. Atrial fibrillation must be detected and anticoagulated. Diabetes control reduces microvascular risk (lacunar strokes). Smoking cessation reduces stroke risk by 50% within 2 years.

10. Stroke Systems of Care in India

India's stroke care faces major challenges: most acute strokes arrive beyond the thrombolysis window; limited stroke unit availability; low awareness of FAST symptoms; inadequate ambulance infrastructure; high out-of-pocket treatment costs.

Stroke units: shown to reduce mortality and dependency. National AIIMS-based stroke registry and programmes like INSPIRE (Indian Stroke Programme Initiative for Research and Education) are building capacity. State-based stroke networks (e.g., Tamil Nadu) are improving access.

Telestroke: remote stroke neurologist consultation via video call enables 24/7 expert support for district hospitals with basic CT capability. A hub-and-spoke model is the most scalable approach for India.

Post-stroke care: only ~20% of Indian stroke patients access formal rehabilitation. Community-based rehabilitation (CBR), family training by OT and PT, and low-cost orthotic devices are critical for the majority who cannot access urban tertiary rehab centres.